



Experiment 9

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Branch: CSE

Semester: 6th

Subject Name: Full Stack Development – II

UID: 23BCS10193

Section/Group: KRG 3-B

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Subject Code: 23CSH-309

1. Aim: To design and implement a Spring Boot application that integrates third-party APIs including **Stripe Payment Gateway** for handling online transactions and **Gmail SMTP Service** for sending email notifications after successful payments.

2. Objective:

- To understand the concept of third-party API integration
- To integrate Stripe Payment Gateway for secure online payments
- To configure and use Gmail SMTP for sending emails
- To implement REST APIs for payment processing
- To handle payment success and failure responses
- To send automated email notifications after successful payment
- To follow Layered Architecture (Controller – Service – Repository)
- To enhance real-world application functionality using external services

3. Implementation / Code:

Tools & Technologies Used:-

- Java
- Spring Boot
- Spring Web
- Spring Data JPA
- Maven
- IntelliJ IDEA
- Stripe API
- Gmail SMTP (Java Mail Sender)
- Postman / Browser
- PostgreSQL Database

Implementation Description:-

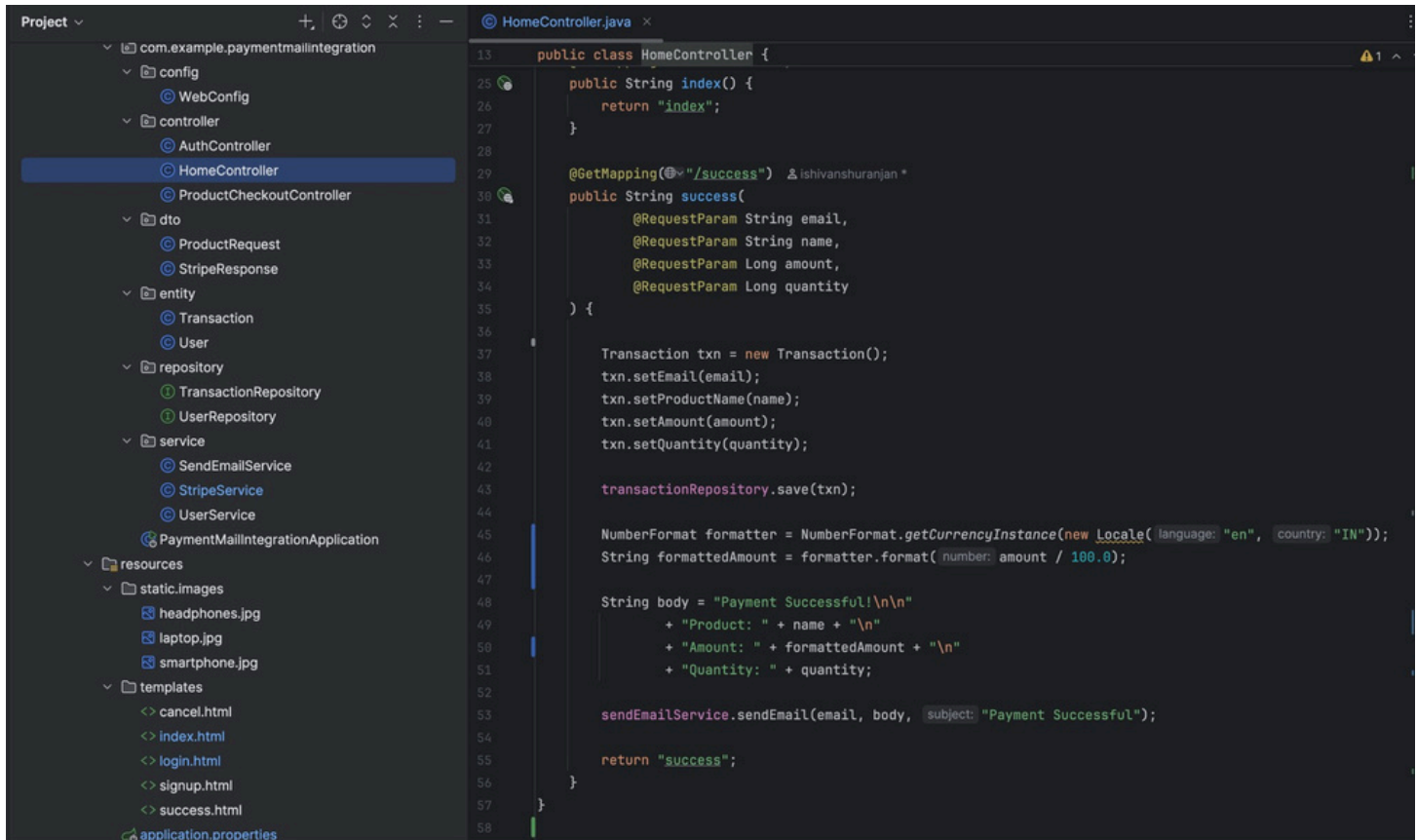
- A Spring Boot project named Payment Mail Integration is created.
- The project integrates Stripe API for handling online payments securely.
- Users enter card details on Stripe Checkout page and complete payment.

After successful payment, a success page is displayed.

- A confirmation email is automatically sent to the user using Gmail SMTP.

Project Structure:-

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4. Output:

- Spring Boot application started successfully on **localhost:8080**
- Stripe payment page loaded successfully
- Payment completed successfully using test card
- Success page displayed after payment completion
- Email notification sent successfully to registered email
- Email contains product details, amount, and quantity
- Payment flow tested successfully using Stripe test environment
- Application integrated successfully with third-party APIs



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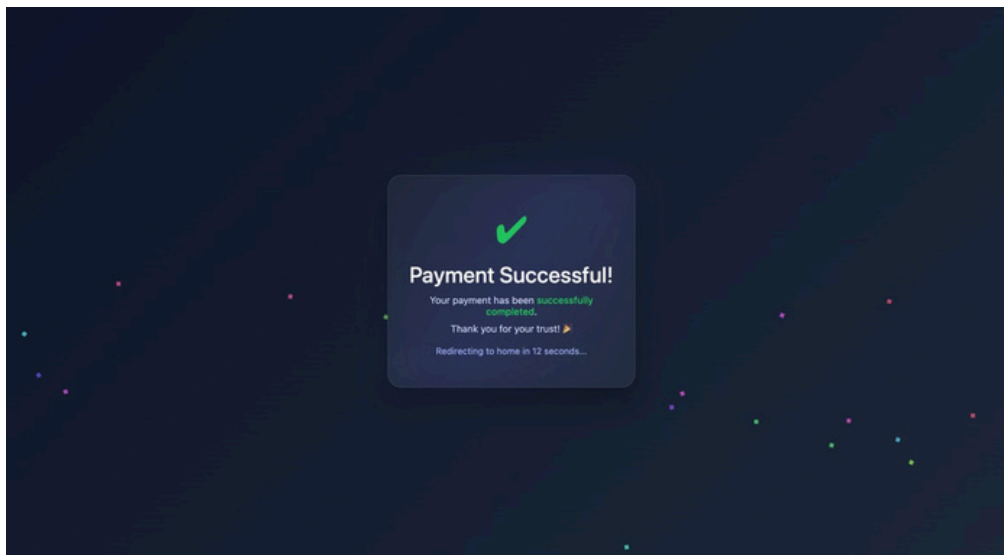
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5. Learning Outcomes (What I Have Learnt):

- Understanding of third-party API integration
- Working with Stripe Payment Gateway
- Secure handling of online transactions
- Sending emails using Gmail SMTP
- Implementing real-world payment systems
- Integration of frontend and backend systems
- Usage of REST APIs in real applications
- Handling success and failure scenarios
- Improving application functionality using external services